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Intro to Probability

⇒ Probability: It's about determining the likelihood of an event.

⇒ Mutual Exclusive event: Two events are mutual exclusive if they can't occur at the same time.
eg: tossing a coin can't give both H & T at same time

⇒ $P_x(H \text{ or } T) = P_x(H) + P_x(T) \rightarrow$ additive rule for mutually exclusive event

⇒ $\{1, 2, 3, 4, 5, 6\}$

$$\therefore P(1 \text{ or } 5) = P(1) + P(5) = 1/6 + 1/6 = 1/3$$

⇒ Non mutual exclusive event: taking a card from deck. we can get K or can get K & ♡

$$\Rightarrow P(K \text{ or } \heartsuit) = P(K) + P(\heartsuit) - P(K \& \heartsuit)$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{8}{26} = \frac{4}{13}$$

additive rule for non mutual exclusive event

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Multⁿ Rule:

⇒ Independent events: if they don't affect one another.
eg. tossing a coin,

⇒ Dep. events: if they affect each other.
eg. take a card from deck & then Queen card,

⇒ Multⁿ rule:

⑤ ind. event:

$$P(H \text{ and } T) = P(H) * P(T)$$

⑥ dep. event:

$$P(K \text{ and } Q) = P(K) * P(Q|K)$$

$$= \frac{4}{52} * \frac{4}{51}$$

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